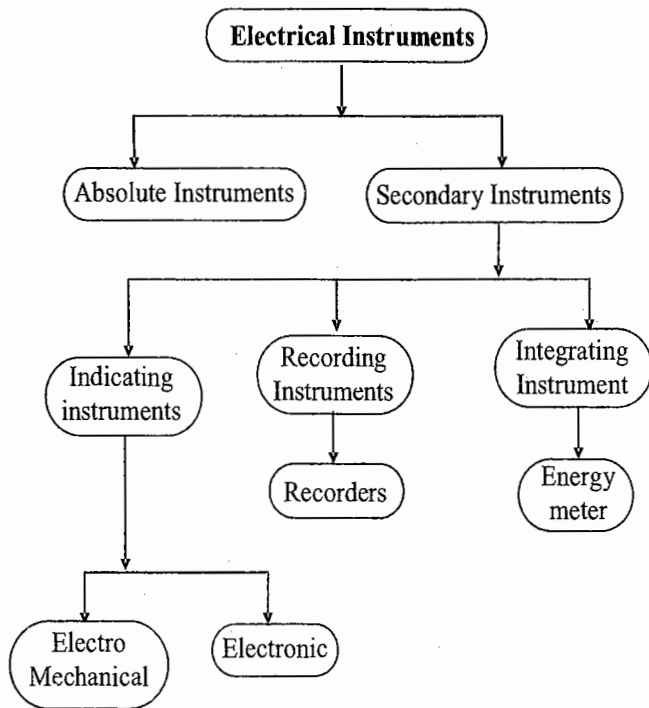


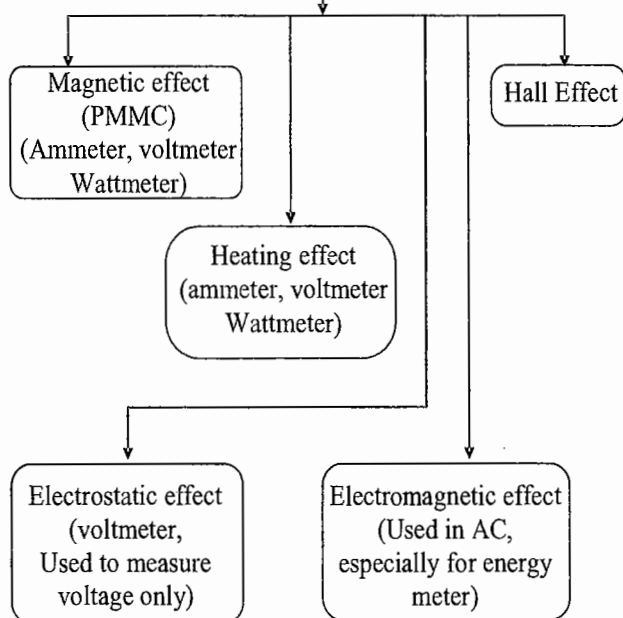
1

Basics of Electrical Instruments

ELECTRICAL INSTRUMENTS:



Principle of operation of indicating instruments



Absolute instrument:

These instruments doesn't give direct readings but gives in terms of instrumental constant. But these are accurate meters. Hence these instruments are used in researching laboratories.

Eg: Tangent galvanometer, Religh's current balance, Absolute electrometer

Secondary instruments: These instruments give direct readings.

Indicating instruments: These instruments will give instantaneous values of the quantity to be measured. Scale and Pointer mechanism are invalid

Eg: PMMC, MI, Dynamometer wattmeter, frequency meter, power factor meter

Recording instrument:

These instruments record the value to be measured (or) observed. Recorded over a graph paper by a light weight pen

- These are used to observe load variation continuously.

Eg: Recording voltmeter, Recording wattmeter, storage oscilloscope

Integrating instruments:

These instruments add the measured value to the existing value. These instruments will give the total electricity consumed over a period of time

Eg: Energy meter (or) kWhr, kVARh meter and ampere hour (Ah) meter

Indicating instruments:

Range: the meter is calibrated to read a value from minimum to maximum value



Span: Its length of the scale. It can be calculated by the algebraic difference of maximum and minimum value

Deflecting torque (T_d):

- Electromagnetic torque (T_{em})
- For the produced deflecting torque the pointer starts from zero position
- If the meter is provided with only deflecting torque, then pointer indicates full value

Controlling torque (T_c):

It is to stop the pointer at the required final value, controlling torque acts exactly equal and in opposite direction to deflecting torque.

Damping torque: The pointer makes oscillations with respect to final value, to eliminate these oscillations damping torque is required.

NOTE: In meter, first produces ' T_d ' proportional to quantity to be measured and then T_c starts producing and raises until it balances the T_d . At final steady state position $T_c = T_d$. So no damping torque (Because speed of the pointer is zero)

NOTE: T_c depends on T_d but damping torque independent of T_c and T_d .

Deflecting torque can be produced by:

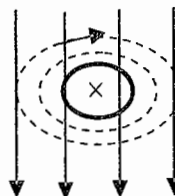
1. Magnetic effect
2. Electrostatic effect
3. Induction effect
4. Thermal effect
5. Chemical effect
6. Hall effect

Magnetic effect:

Whenever a current carrying conductor placed in magnetic field it experiences force by **Lorentz's** principle

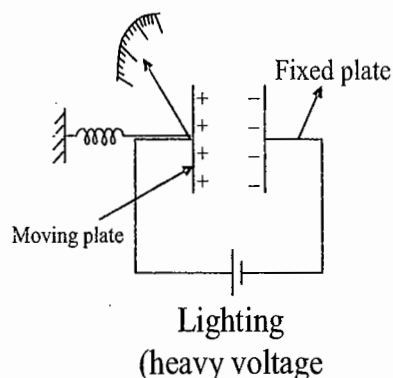
$$F = BIL\sin\theta$$

That force direction is found by **fleming left hand rule**



Here conductor moves "right to left"
(By fleming's left hand rule)

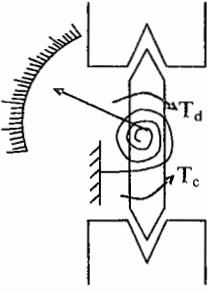
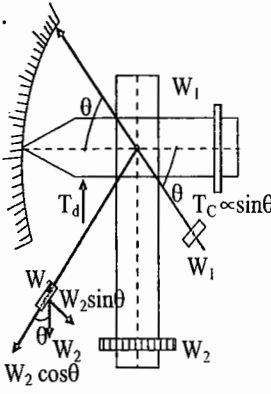
Electrostatic effect:



- When a high voltage is applied to capacitor then capacitor plates are charged with opposite polarities electro static force is produced, because of this the plates will move together.

If one plate is fixed, pointer is connected to movable plate Then we can see the voltage measured by pointer. These are specially used to measure very high dc voltages.

Controlling torque:

Spring control	Gravity control
1. Hair spring is used to provide the control	1. Weights are used to balance the motion of pointer
2. 	2. 
3. Springs are connected, in series with moving coil i.e. Springs are carrying current into the coil and out from the coil	3. These must be placed in vertical direction only
4. In MI instrument there is no physical contact between coil and spring i.e spring is not carrying the current	4. This method will give non linear scales
5. Spring is made up of phosphorous bronze	5. Controlling torque can be only upto 90°. After this controlling torque will decreases.

Note: In PMMC spring snaps means it's broken so no current it's carrying

Note: Spring control instruments can placed in any direction and gives linear scales. Hence it is used in all instruments

Example 1:

A weight of a 3.5g is used as the controlling weight in a gravity controlled instrument. Find its distance from spindle if the deflecting torque corresponding to a deflection of 58° is $2.51 \times 10^{-3} \text{ Nm}$

- (a) 45.2mm (b) 78.6 mm
(c) 66.3 mm (d) 86.2mm

Ans:(d)

Sol: Deflecting torque due to a weight is

$$T_d = mg \sin \theta \times d$$

Where,

d = distance of the weight from spindle

$$d = \frac{T_d}{mg \sin \theta}$$

$$= \frac{2.51 \times 10^{-3}}{3.5 \times 10^{-3} \times 9.81 \times \sin 58^\circ}$$

$$= 0.0862 \text{ m} = 86.2 \text{ mm}$$

Eddy current damping:

- In this method Al vane and permanent magnet are used
- Most effective damping method
- This method is suitable for the instruments having strong magnetic field (i.e. permanent magnet)

Eg: PMMC

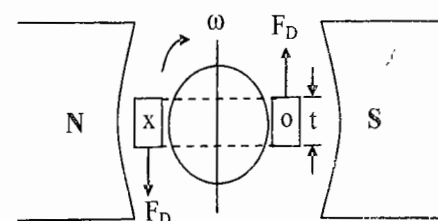


Fig: Eddy current damping by metal former

Air friction damping:

The gap between Al piston and cylinder hole must be very small and should be uniform.

- This method is suitable for the instrument having weak magnetic field

Ex: MI Instrument, Electro dynamo meter, Watt meter.

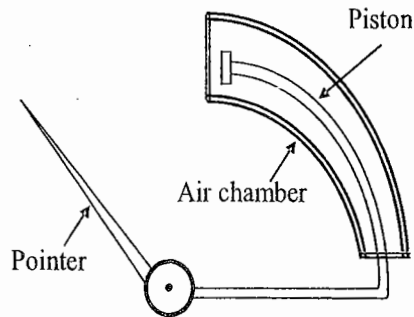


Fig: Air friction damping

Fluid friction damping:

Here high viscous fluid is used (mineral oil) used for instrument having low deflecting torque

Eg: Electro static voltmeter

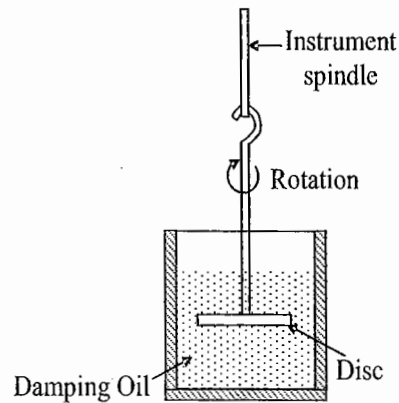


Fig: Fluid Friction Damping

These instruments must be placed in vertical direction only

- These are not portable instruments

Effectiveness order:

Eddy current > fluid friction > Air friction

- Under damping oscillations are present and pointer takes more time to give the reading
- For over damping no oscillations but pointer moves very slowly to the setting.
- For critical damping pointer reaches final value in quickest possible time without any oscillations called "dead beat"
- All indicating instruments are 2nd order instruments i.e. $0.6 < \xi < 0.8$. Here time taken by pointer depends only on "damping torque."